

SuRaksha Cyber Hackathon 2.0

Theme 1 — Real-Time Anomaly Detection

Idea Phase Submission Document

Project Title

SentinelX — Real-Time Banking Fraud & Behavioral Anomaly Detection System

Team Details

Team Member	Role	Responsibilities
Ganji Lokesh	Cybersecurity & Detection Logic	Fraud scenarios, anomaly rules, security workflows
Badimela Sunil	Backend Developer	APIs, database integration, authentication engine
Munagapati Tenesh Sai Kamal	Frontend & Integration Developer	Dashboard, alerts UI, frontend-backend integration, deployment

1. Executive Summary

Digital banking adoption in India has increased rapidly over the last few years. Along with this growth, financial fraud, account takeovers, document tampering, and identity theft attacks have also increased significantly.

Traditional banking security systems mostly rely on:

- Passwords
- OTP verification
- Manual document verification
- Delayed fraud investigation

These methods are no longer sufficient against modern fraud attacks.

Our proposed solution, SentinelX, is a real-time anomaly detection and behavioral monitoring platform designed specifically for fintech and banking environments.

The system continuously monitors:

- User login behavior
- Device changes
- IP reputation
- Geographic anomalies
- Suspicious access attempts
- Document metadata inconsistencies
- Forgery indicators

The platform assigns dynamic risk scores to activities and alerts bank officers instantly when suspicious behavior is detected.

Our goal is to reduce fraud detection time from several hours to less than 5 minutes while improving underwriting reliability and customer security.

2. Problem Statement

Indian banks are increasingly facing sophisticated cyber fraud attacks involving:

- Fake salary slips
- Forged land records
- Manipulated financial statements
- Unauthorized account access
- Credential theft
- Identity impersonation
- Account takeover attacks

Current fraud detection systems suffer from several limitations:

Existing Problem	Impact
Delayed fraud detection	Financial losses increase rapidly
Manual verification processes	Slow loan approval and underwriting
Weak password-based authentication	Easy target for phishing and credential theft
Lack of behavioral monitoring	Fraudulent activities go undetected
No real-time anomaly scoring	Delayed response from security teams

Many current systems only detect fraud AFTER it has already occurred, Resulting in significant losses.

Banks require:

- Real-time monitoring
 - Automated fraud detection
 - Intelligent behavioral analysis
 - Secure authentication workflows
 - Explainable alerts for bank officers
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3. Proposed Solution

SentinelX is an AI-assisted real-time fraud monitoring and anomaly detection system for banking platforms.

The system uses behavior-based security analysis combined with rule-based fraud detection techniques.

The platform monitors:

- Login patterns
- Device fingerprint changes
- Geolocation anomalies
- Failed authentication attempts
- Transaction irregularities
- Document tampering indicators
- Metadata inconsistencies

When suspicious activity is detected: 1. Risk score is generated 2. Alert is triggered 3. OTP/MFA verification is requested 4. Security dashboard updates instantly 5. Bank officer receives actionable insights

The platform also includes explainable AI concepts so bank staff can understand why a user or document was flagged.

4. Core Innovation

Most traditional fraud systems rely only on static rule-based verification.

Our approach combines:

- Behavioral analysis
- Real-time anomaly scoring
- Device intelligence
- Explainable AI concepts
- Risk-based authentication

Key Innovation Points

1. Multi-Layer Behavioral Monitoring

Instead of checking only passwords, the system monitors:

- Login time patterns
- Device changes
- IP changes
- Location anomalies
- Typing behavior simulation
- Access frequency

2. Dynamic Risk Scoring Engine

Each event receives a live risk score.

Example:

Activity	Risk
Normal login	Low
New device login	Medium
Different country + failed attempts	High
Modified document metadata	Critical

3. Real-Time Alerting

The system alerts bank officers instantly instead of delayed investigation.

4. Explainable Security Alerts

The dashboard explains:

- Why activity was flagged
- Which anomaly triggered detection
- Suggested security action

This improves trust and operational efficiency.

We will use LIME (Local Interpretable Model-agnostic Explanations) to explain why each transaction was flagged. For example, the dashboard will show:

High fraud risk (85%) because:

- Device changed from iPhone to Unknown Device (35% weight)
 - Login from unusual location: USA vs usual location India (40% weight)
 - Failed login attempts: 3 in past hour (25% weight)""
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5. Objectives

The main objectives of SentinelX are:

- Detect fraud attempts in real time
 - Reduce manual verification workload
 - Improve customer account protection
 - Detect document tampering attempts
 - Provide intelligent underwriting support
 - Minimize financial losses
 - Strengthen digital trust in banking systems
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6. Features

Authentication & Login Monitoring

- Device fingerprint tracking
- Login location analysis
- IP reputation analysis
- Failed login monitoring
- Behavioral anomaly detection

Fraud Detection Engine

- Dynamic risk scoring
- Suspicious behavior classification
- Real-time fraud alert generation
- Rule-based anomaly engine

Security Dashboard

- Live monitoring interface
- Fraud alert visualization
- Risk analytics charts

- Incident timeline tracking

Alert Response Workflow

- OTP verification trigger
 - Admin escalation mechanism
 - Account lockdown option
 - Alert prioritization
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7. Real-World Use Case

Scenario: Fake Loan Application Fraud

A user uploads:

- Salary slip
- Financial statement
- Land document

The system detects:

- Suspicious metadata edits
- Device change anomaly
- Login from unusual location
- Multiple failed login attempts

SentinelX generates:

Risk Score: 92/100

Threat Level: HIGH

Recommended Action: Manual Verification Required

Bank officer receives instant alert before loan approval.

This reduces fraud risk significantly.

8. System Architecture

User Activity / Document Upload



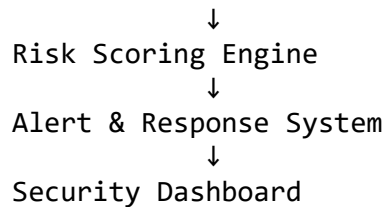
Data Collection Layer



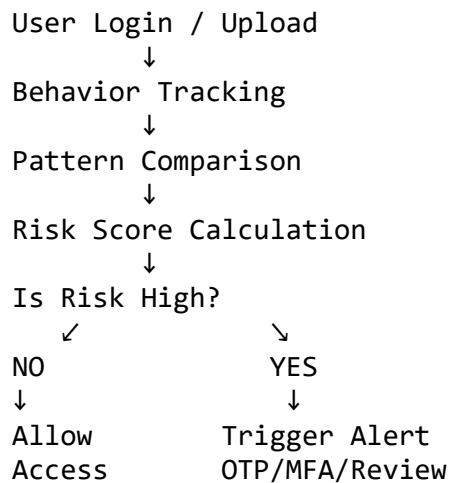
Behavior Monitoring Engine



Anomaly Detection Module



9. Workflow Diagram



10. Technology Stack

Layer	Technology
Backend	Python 3.11 + Flask
Frontend	React.js + Tailwind CSS
Database	PostgreSQL
ML & Anomaly Detection	scikit-learn (Isolation Forest)
Data Visualization	Plotly
Explainability	LIME (Local Interpretable Model-agnostic Explanations)
Authentication	JWT tokens + OTP APIs
Deployment	Heroku
Version Control	GitHub

11. Why Our Tech Stack Is Realistic

Our stack is intentionally simple and practical.

We avoided:

- Overly complex enterprise tools
- Unnecessary blockchain integration
- Multiple cloud providers
- Heavy infrastructure

Reason: Hackathon timeline is limited.

The selected stack allows:

- Faster development
- Easier debugging
- Better team collaboration
- Faster deployment
- Working prototype within timeline

Why Flask Over FastAPI:

- ✓ FastAPI is powerful but heavier
- ✓ Flask is lightweight and faster to learn
- ✓ Perfect for 4-week hackathon MVP
- ✓ Flask + Plotly = instant interactive dashboard

Why Plotly for Data Visualization:

- ✓ Interactive charts (real-time updates)
- ✓ Easy to integrate with Flask
- ✓ Professional looking fraud dashboards
- ✓ No separate frontend needed

Why Heroku for Deployment:

- ✓ Free tier available
- ✓ No infrastructure setup needed
- ✓ One-click deployment
- ✓ Automatic SSL certificates

12. Data Strategy

Real banking datasets are confidential.

Therefore, we will use:

Public & Synthetic Data Sources

Synthetic Data

Generated using:

- Python Faker library
- Simulated banking logs
- Simulated login behavior patterns

Open Security Datasets

Possible datasets:

- CSIC 2010 HTTP Dataset
- Credit Card Fraud Detection Dataset
- Kaggle Fraud Detection datasets

This approach ensures:

- Privacy compliance
 - Faster experimentation
 - Safe testing environment
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13. Security & Privacy Considerations

Since banking systems handle sensitive information, security and compliance are critical.

Security Measures

Security Layer	Implementation
Data Encryption	AES-256 encryption
Authentication	JWT + OTP verification
Password Security	Hashed passwords
Session Security	Session timeout + token validation
Logging	Secure audit logs

Privacy Measures

- No raw password storage
- Sensitive data masking
- Minimal data retention
- Role-based dashboard access

Compliance Awareness

The system design considers:

- RBI digital banking guidelines
 - Data protection principles
 - Secure audit logging practices
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14. Risk Analysis & Mitigation

Risk	Impact	Mitigation
Lack of real banking data	Model limitations	Use synthetic datasets
False positives	User inconvenience	Threshold-based validation
Overfitting	Poor detection quality	Cross-validation techniques
API integration issues	Delayed development	Modular backend design
Limited hackathon time	Feature reduction	Prioritize MVP first

15. Feasibility Analysis

The proposed system is achievable within hackathon timeline because:

- Scope is clearly limited
- Core features are prioritized
- Stack is lightweight
- Team roles are divided efficiently
- Prototype-focused development strategy

We are NOT attempting:

- Full banking platform
- Production-scale deployment
- Enterprise-grade infrastructure

We are building:

- A functional proof-of-concept prototype
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16. Minimum Viable Product (MVP)

The hackathon prototype version of SentinelX will focus on delivering the core functionalities required for real-time fraud monitoring and anomaly detection within the given 4-week implementation timeline.

The MVP will include:

- Login behavior monitoring
- Device anomaly detection
- Real-time risk scoring
- Fraud alert dashboard
- OTP trigger for suspicious logins

The MVP prioritizes:

- functionality,
- practical implementation,
- real-time monitoring,
- realistic deployment scope.

Advanced enterprise-scale AI features such as deep learning-based behavioral biometrics and large-scale banking integrations are planned as future enhancements beyond the hackathon prototype stage.

17. Development Roadmap

Week 1 — Research & System Design

- Research banking fraud patterns
 - Finalize project architecture
 - Create synthetic datasets
 - Setup GitHub repository
 - Define anomaly detection rules
 - Design database schema
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Week 2 — Backend & Detection Engine Development

- Build backend APIs using Flask

- Implement login tracking module
 - Create anomaly detection logic
 - Develop dynamic risk scoring engine
 - Integrate database connectivity
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Week 3 — Frontend Dashboard & Integration

- Build admin monitoring dashboard
 - Integrate frontend with backend APIs
 - Add real-time fraud alert system
 - Implement OTP/MFA workflow
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Week 4 — Testing, Optimization & Final Submission

- Comprehensive testing & debugging
 - Performance optimization
 - UI/UX refinement
 - Final documentation
 - Demo video recording
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18. Expected Deliverables

At the end of the hackathon, Team SentinelX will submit the following deliverables:

Deliverable 1 — Functional Web-Based Prototype

A working prototype capable of:

- monitoring login behavior,
- detecting suspicious activity,
- assigning dynamic risk scores,
- generating real-time fraud alerts.

Deliverable 2 — Real-Time Security Monitoring Dashboard

An interactive admin dashboard displaying:

- live fraud alerts,
- high-risk login attempts,
- anomaly statistics,

- device and location tracking,
- incident timelines.

Deliverable 3 — Risk Scoring & Anomaly Detection Engine

A backend detection engine implementing:

- behavioral anomaly rules,
- suspicious login detection,
- device mismatch analysis,
- failed login tracking,
- rule-based fraud scoring.

Deliverable 4 — Backend APIs & Database Integration

A secure backend system including:

- authentication APIs,
- risk-analysis APIs,
- PostgreSQL database integration,
- secure logging workflows.

Deliverable 5 — GitHub Repository

A public/private GitHub repository containing:

- complete source code,
- project structure,
- setup instructions,
- API documentation,
- commit history,
- development workflow.

Deliverable 6 — Technical Documentation

Comprehensive documentation including:

- architecture diagrams,
- workflow diagrams,
- feature explanations,
- technology stack details,
- security considerations,
- implementation methodology.

Deliverable 7 — Demo Video Presentation

A 5-minute demonstration video showcasing:

- project overview,
- real-time fraud detection workflow,
- dashboard functionality,
- alert generation,
- system architecture,
- practical banking use case simulation.

Deliverable 8 — Final Presentation Deck

A professional presentation covering:

- problem statement,
 - market relevance,
 - innovation,
 - implementation strategy,
 - real-world impact,
 - scalability,
 - future enhancements.
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19. Competitive Advantage

Existing Systems	SentinelX
Mostly password-based	Behavior-based monitoring
Delayed fraud detection	Real-time alerts
Manual verification	Automated anomaly scoring
Static rules	Adaptive risk scoring
Limited explainability	Transparent alert reasoning

20. Real-World Impact

If implemented in banking systems:

- Fraud detection time can reduce significantly
- Underwriting reliability improves
- Customer trust increases
- Manual workload decreases

- Financial losses reduce

Potential Impact

For a bank with:

- 1 million customers
- thousands of daily transactions

Even small fraud reduction percentages can save crores annually.

The system is scalable and can later integrate with:

- UPI systems
 - Internet banking
 - Loan processing systems
 - Fraud monitoring centers
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21. Future Scope

Future enhancements may include:

- Deep learning-based anomaly detection
 - Facial recognition authentication
 - Behavioral biometrics
 - SIEM integration
 - Mobile app monitoring
 - Threat intelligence feeds
 - AI-powered investigation assistant
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22. Why This Solution Matters

Digital banking security is becoming one of the most critical challenges in India.

Current systems are reactive.

SentinelX aims to make fraud detection:

- proactive
- intelligent
- explainable
- real-time

Our system combines cybersecurity thinking, anomaly detection concepts, and practical fintech security workflows into one unified platform.

23 Conclusion

SentinelX is a practical, scalable, and security-focused anomaly detection platform designed to strengthen banking fraud prevention.

By combining:

- behavioral monitoring
- risk scoring
- document verification
- real-time alerting

our solution helps banks detect suspicious activity faster and improve digital trust.

The project aligns strongly with the SuRaksha Hackathon Theme 1 challenge and addresses real-world fintech security problems using realistic technologies and implementation strategies.

24. GitHub & Project Repository

GitHub Repository:

<https://github.com/Lokesh742/sentinelx>

25. Appendix — Example Fraud Detection Rules

Rule	Risk Level
Login from unknown country	High
Multiple failed logins	Medium
Device fingerprint mismatch	High
Metadata modification detected	Critical
Impossible travel login	Critical
Repeated OTP failures	High

26. Appendix — Sample Dashboard Metrics

The dashboard will display:

- Total fraud alerts
 - High-risk users
 - Failed login attempts
 - Device anomalies
 - Tampered document count
 - Real-time risk heatmap
 - Incident timelines
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27. Final Statement

Our team believes cybersecurity in fintech should not only be reactive but predictive and intelligent.

SentinelX represents a realistic step toward smarter digital banking protection systems that are scalable, explainable, and operationally practical.